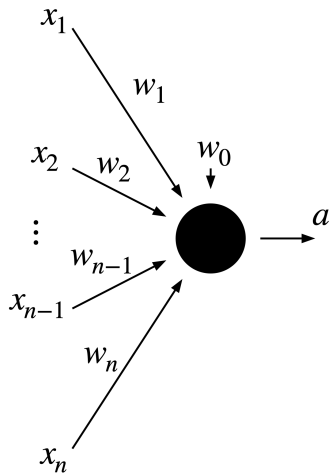




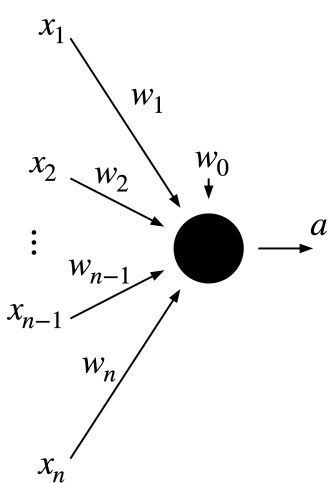
Feedforward Artificial Neural Networks (FANN)

Alejandro Veloz

Single neuron operations



Single neuron operations

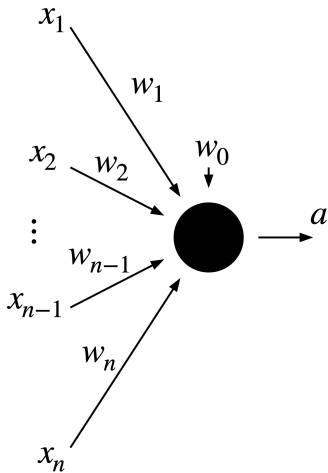


- It receives **input data point**: $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$
- and computes the **output**:

$$\underbrace{a}_{\text{activation}} = \sigma \left(\underbrace{z}_{\text{preactivation}} \right) = \sigma \left(\underbrace{\sum_{j=1}^n x_j w_j + w_0}_z \right)$$

- $z = \sum_{j=1}^n x_j w_j + w_0$ is called **preactivation**.
- $\sigma(z)$ is the **activation function**.
- $\mathbf{w} = [w_1, \dots, w_n]^\top$ are **weights** (parameters).
- w_0 is the **bias parameters**.

Single neuron operations

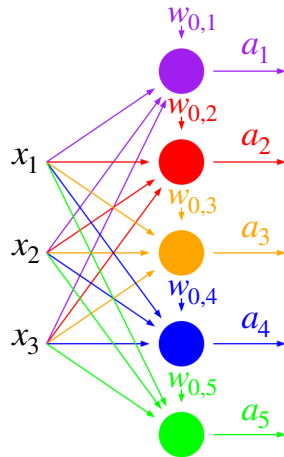


$$\begin{aligned} a &= \sigma \left(\underbrace{\sum_{j=1}^n x_j w_j + w_0}_z \right) \\ &= \sigma \left(\underbrace{\begin{bmatrix} w_1 & \dots & w_n \end{bmatrix}}_{\mathbf{w}^\top} \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{\mathbf{x}} + w_0 \right) \\ &= \sigma (\mathbf{w}^\top \mathbf{x} + w_0) \end{aligned}$$

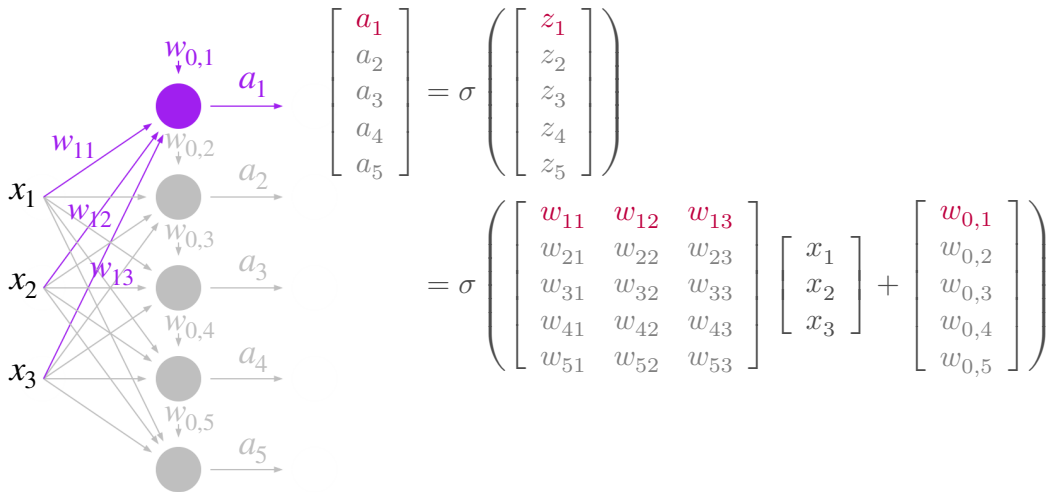
Single layer operations

Consider 5 neurons that operates "in parallel".

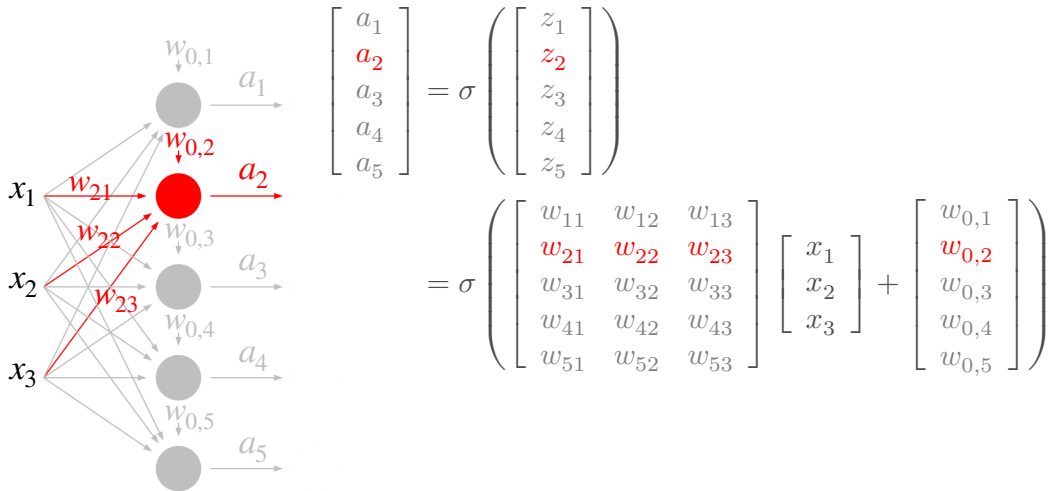
It receives **input** x and computes the **output** a



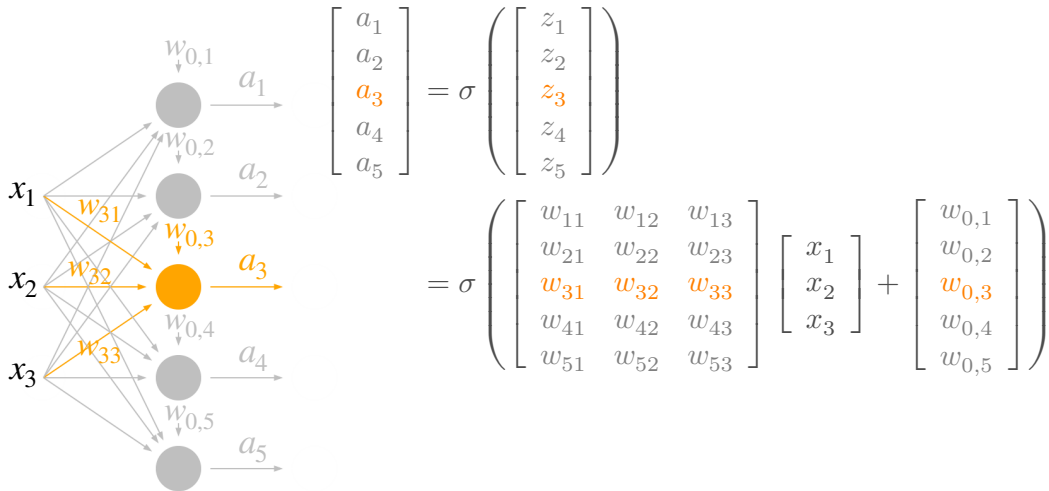
Single layer operations



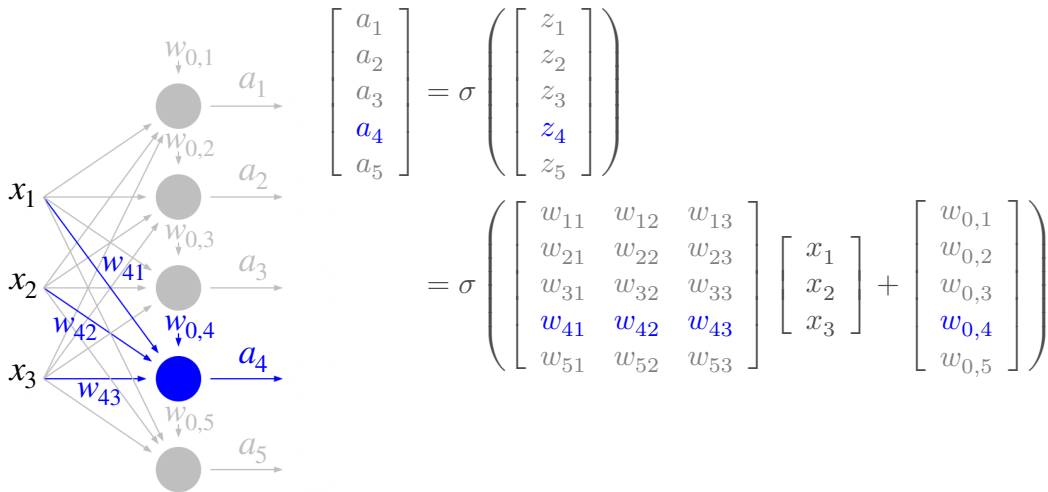
Single layer operations



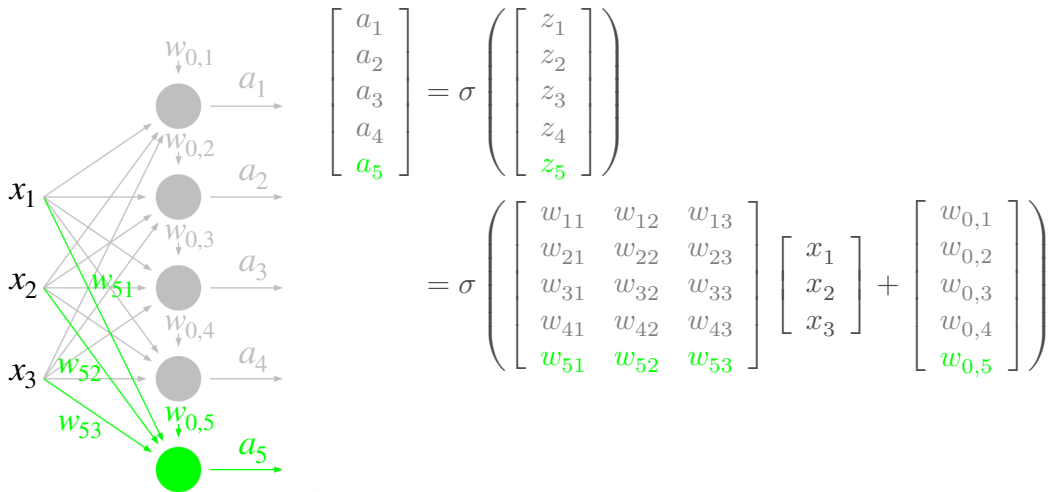
Single layer operations



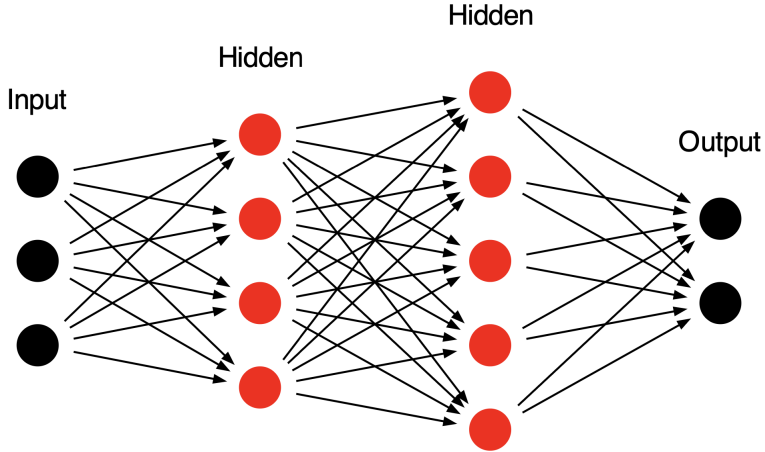
Single layer operations



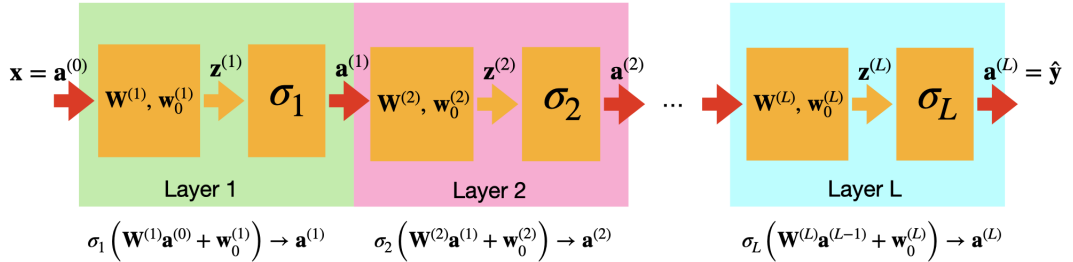
Single layer operations



Network with multiple layers



Network with multiple layers



Single layer operations

Consider a network of L layers (one input layer and $L - 1$ layers of neurons).

Each layer is composed by $n^{(\ell)}$ neurons, $\ell \in \{0, 1, \dots, L\}$.

The input vector for layer ℓ is:

$$\mathbf{a}^{(\ell-1)} = \begin{bmatrix} a_1^{(\ell-1)} \\ \vdots \\ a_{n^{(\ell-1)}}^{(\ell-1)} \end{bmatrix}$$

The output vector for layer ℓ is:

$$\mathbf{a}^{(\ell)} = \begin{bmatrix} a_1^{(\ell)} \\ \vdots \\ a_{n^{(\ell)}}^{(\ell)} \end{bmatrix}$$

Single layer operations

$$\underbrace{\begin{bmatrix} a_1^{(\ell)} \\ \vdots \\ a_{n^{(\ell)}}^{(\ell)} \end{bmatrix}}_{\mathbf{a}^{(\ell)}} = \sigma_{\ell} \left(\begin{bmatrix} z_1^{(\ell)} \\ \vdots \\ z_{n^{(\ell)}}^{(\ell)} \end{bmatrix} \right)$$
$$= \sigma_{\ell} \left(\begin{bmatrix} \mathbf{w}_1^{(\ell)\top} \\ \vdots \\ \mathbf{w}_{n^{(\ell)}}^{(\ell)\top} \end{bmatrix} \begin{bmatrix} a_1^{(\ell-1)} \\ \vdots \\ a_{n^{(\ell-1)}}^{(\ell-1)} \end{bmatrix} + \begin{bmatrix} w_{0,1}^{(\ell)} \\ \vdots \\ w_{0,n^{(\ell)}}^{(\ell)} \end{bmatrix} \right) = \sigma \left(\mathbf{W}^{(\ell)} \mathbf{a}^{(\ell-1)} + \mathbf{w}_0^{(\ell)} \right)$$

Feedforward network operations

Given the input vector $\mathbf{x}$ and a network of L layers:

Layer 1: $\mathbf{a}^{(1)} = \sigma_1 \left(\mathbf{W}^{(1)} \mathbf{x} + \mathbf{w}_0^{(1)} \right)$

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$\vdots$

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$\vdots$

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$\vdots$

Layer L : $\hat{\mathbf{y}} = \mathbf{a}^{(L)} = \sigma_L \left(\mathbf{W}^{(L)} \mathbf{a}^{(L-1)} + \mathbf{w}_0^{(L)} \right)$

Feedforward network operations

By defining $\mathbf{a}^{(0)} = \mathbf{x}$:

Layer 1: $\mathbf{a}^{(1)} = \sigma_1 \left(\mathbf{W}^{(1)} \mathbf{a}^{(0)} + \mathbf{w}_0^{(1)} \right)$

Layer 2: $\mathbf{a}^{(2)} = \sigma_2 \left(\mathbf{W}^{(2)} \mathbf{a}^{(1)} + \mathbf{w}_0^{(2)} \right)$

$\vdots$

Layer ℓ : $\mathbf{a}^{(\ell)} = \sigma_\ell \left(\mathbf{W}^{(\ell)} \mathbf{a}^{(\ell-1)} + \mathbf{w}_0^{(\ell)} \right)$

$\vdots$

Layer L : $\mathbf{a}^{(L)} = \sigma_L \left(\mathbf{W}^{(L)} \mathbf{a}^{(L-1)} + \mathbf{w}_0^{(L)} \right)$

Learning the network parameters: Backpropagation

Training process

Two steps:

1. **Backward Pass (Backpropagation of Error):** Compute the derivatives of the error function with respect to the weights by propagating errors backward through the network.
2. **Weight Update:** Adjust the weights using the calculated derivatives, typically by applying an optimization method such as gradient descent.

Training process

- Training dataset: $\{(\mathbf{x}_1, \mathbf{y}_1), \dots, (\mathbf{x}_D, \mathbf{y}_D)\}$

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- Loss function $\text{loss}(\text{guess}, \text{actual})$:

$$J(\mathbf{W}, \mathbf{W}_0) = \sum_{d=1}^D \text{loss}(\text{NN}(\mathbf{x}_d; \mathbf{W}, \mathbf{W}_0), \mathbf{y}_d),$$

where NN is the output of our neural network for a given input.

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where NN is the output of our neural network for a given input.

- We can do (stochastic/batch) gradient descent, adjusting the weights $\mathbf{W}$, $\mathbf{W}_0$ to minimize J .

Define $\hat{y} = \text{NN}(\mathbf{x}; \mathbf{W}, \mathbf{W}_0)$ the network response under the data point $\mathbf{x}$.

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Gradient descent rule:

$$\mathbf{W}(t+1) = \mathbf{W}(t) - \eta \nabla_{\mathbf{W}} \text{loss}(\hat{y}, \mathbf{y})$$

$$\mathbf{W}_0(t+1) = \mathbf{W}_0(t) - \eta \nabla_{\mathbf{W}_0} \text{loss}(\hat{y}, \mathbf{y})$$

Recall:

$$\mathbf{a}^{(\ell)} = \sigma_{\ell}(\mathbf{z}^{(\ell)}) \quad \mathbf{z}^{(\ell)} = \mathbf{W}^{(\ell)}\mathbf{a}^{(\ell-1)} + \mathbf{w}_0^{(\ell)}$$

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The derivative of the loss w.r.t. $\mathbf{W}^{(L)}$ is:

$$\frac{\partial \text{loss}}{\partial \mathbf{W}^{(L)}} = \underbrace{\frac{\partial \text{loss}}{\partial \mathbf{a}^{(L)}}}_{\text{depends on loss function}} \cdot \underbrace{\frac{\partial \mathbf{a}^{(L)}}{\partial \mathbf{z}^{(L)}}}_{\sigma'_L(\mathbf{z}^{(L)})} \cdot \underbrace{\frac{\partial \mathbf{z}^{(L)}}{\partial \mathbf{W}^{(L)}}}_{\mathbf{a}^{(L-1)}}$$

$$\frac{\partial \text{loss}}{\partial \mathbf{W}^{(L)}} = \frac{\partial \text{loss}}{\partial \mathbf{a}^{(L)}} \frac{\partial \mathbf{a}^{(L)}}{\partial \mathbf{z}^{(L)}} \frac{\partial \mathbf{z}^{(L)}}{\partial \mathbf{W}^{(L)}}$$

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$$\frac{\partial \text{loss}}{\partial \mathbf{W}^{(L-1)}} = \frac{\partial \text{loss}}{\partial \mathbf{a}^{(L)}} \frac{\partial \mathbf{a}^{(L)}}{\partial \mathbf{z}^{(L)}} \frac{\partial \mathbf{z}^{(L)}}{\partial \mathbf{a}^{(L-1)}} \frac{\partial \mathbf{a}^{(L-1)}}{\partial \mathbf{z}^{(L-1)}} \frac{\partial \mathbf{z}^{(L-1)}}{\partial \mathbf{W}^{(L-1)}}$$

$$\vdots$$

$$\frac{\partial \text{loss}}{\partial \mathbf{W}^{(L)}} = \frac{\partial \text{loss}}{\partial \mathbf{a}^{(L)}} \frac{\partial \mathbf{a}^{(L)}}{\partial \mathbf{z}^{(L)}} \frac{\partial \mathbf{z}^{(L)}}{\partial \mathbf{W}^{(L)}}$$

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$$\vdots$$

$$\frac{\partial \text{loss}}{\partial \mathbf{W}^{(L)}} = \frac{\partial \text{loss}}{\partial \mathbf{a}^{(L)}} \sigma'_L(\mathbf{z}^{(L)}) \mathbf{a}^{(L-1)}$$

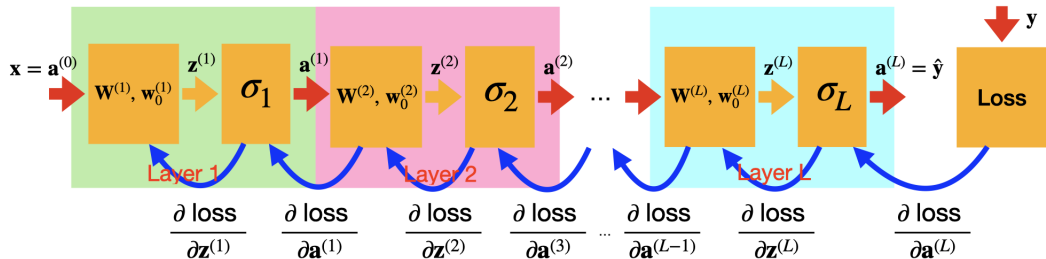
$$\frac{\partial \text{loss}}{\partial \mathbf{W}^{(L-1)}} = \frac{\partial \text{loss}}{\partial \mathbf{a}^{(L)}} \sigma'_L(\mathbf{z}^{(L)}) \mathbf{W}^{(L)} \sigma'_{L-1}(\mathbf{z}^{(L-1)}) \mathbf{a}^{(L-2)}$$

$$\vdots$$

The general case

$$\begin{aligned}\frac{\partial \text{loss}}{\partial \mathbf{W}^{(\ell)}} &= \frac{\partial \text{loss}}{\partial \mathbf{a}^{(L)}} \frac{\partial \mathbf{a}^{(L)}}{\partial \mathbf{z}^{(L)}} \frac{\partial \mathbf{z}^{(L)}}{\partial \mathbf{a}^{(L-1)}} \frac{\partial \mathbf{a}^{(L-1)}}{\partial \mathbf{z}^{(L-1)}} \frac{\partial \mathbf{z}^{(L-1)}}{\partial \mathbf{a}^{(L-2)}} \frac{\partial \mathbf{a}^{(L-2)}}{\partial \mathbf{z}^{(L-2)}} \\ &\quad \cdots \frac{\partial \mathbf{z}^{(\ell+1)}}{\partial \mathbf{a}^{(\ell)}} \frac{\partial \mathbf{a}^{(\ell)}}{\partial \mathbf{z}^{(\ell)}} \frac{\partial \mathbf{z}^{(\ell)}}{\partial \mathbf{W}^{(\ell)}} \\ &= \frac{\partial \text{loss}}{\partial \mathbf{a}^{(L)}} \sigma'_L \mathbf{W}^{(L)} \sigma'_{L-1} \mathbf{W}^{(L-1)} \sigma'_{L-2} \mathbf{W}^{(L-2)} \\ &\quad \cdots \mathbf{W}^{(\ell+1)} \sigma'_\ell \mathbf{a}^{(\ell-1)} \\ &= \frac{\partial \text{loss}}{\partial \mathbf{z}^{(\ell)}} \mathbf{a}^{(\ell-1)}\end{aligned}$$

Error backpropagation



Backpropagation

```
1: for  $l = 1$  to  $L$  do
2:    $w_{ij}^{(l)} \sim \text{Gaussian}(0, 1/m^l)$  ;     $w_{0j}^{(l)} \sim \text{Gaussian}(0, 1)$ 
3: end for
4: for  $t = 1$  to  $T$  do
5:    $i = \text{random sample from } \{1, \dots, n\}$  , then define  $\mathbf{a}^0 = \mathbf{x}^{(i)}$ 
6:   for  $l = 1$  to  $L$  do
7:      $\mathbf{z}^{(l)} = \mathbf{W}^{l\top} \mathbf{a}^{(l-1)} + \mathbf{W}_0^{(l)}$  ;     $\mathbf{a}^{(l)} = \sigma^{(l)}(\mathbf{z}^{(l)})$ 
8:   end for
9:    $\text{loss} = \text{Loss}(\mathbf{a}^{(L)}, \mathbf{y}^{(i)})$ 
10:  for  $l = L$  to  $1$  do
11:     $\partial \text{loss} / \partial \mathbf{a}^{(l)} = \text{if } l < L \text{ then } \partial \text{loss} / \partial \mathbf{z}^{(l+1)} \cdot \partial \mathbf{z}^{(l+1)} / \partial \mathbf{a}^{(l)} \text{ else } \partial \text{loss} / \partial \mathbf{a}^{(L)}$ 
12:     $\partial \text{loss} / \partial \mathbf{z}^{(l)} = \partial \text{loss} / \partial \mathbf{a}^{(l)} \cdot \partial \mathbf{a}^{(l)} / \partial \mathbf{z}^{(l)}$ 
13:     $\partial \text{loss} / \partial \mathbf{W}^{(l)} = \partial \text{loss} / \partial \mathbf{z}^{(l)} \partial \mathbf{z}^{(l)} / \partial \mathbf{W}^{(l)}$  ,  $\partial \text{loss} / \partial \mathbf{W}_0^{(l)} = \partial \text{loss} / \partial \mathbf{z}^{(l)} \partial \mathbf{z}^{(l)} / \partial \mathbf{W}_0^{(l)}$ 
14:     $\mathbf{W}^{(l)} = \mathbf{W}^{(l)} - \eta(t) \partial \text{loss} / \partial \mathbf{W}^{(l)}$  ,  $\mathbf{W}_0^{(l)} = \mathbf{W}_0^{(l)} - \eta(t) \partial \text{loss} / \partial \mathbf{W}_0^{(l)}$ 
15:  end for
16: end for
17: return  $\mathbf{W}, \mathbf{W}_0$ 
```